

▼ Exploratory Data Analysis (EDA) - Titanic Dataset

Objective: Extract insights using visual and statistical exploration to identify factors affecting survival on the Titanic.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

Titanic = pd.read_csv("https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv")
Titanic.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
Futrelle, Mrs. Jacques Heath (Lilv												

Next steps:

Generate code with Titanic

View recommended plots

New interactive sheet

▼ Data Exploration

```
print("Dataset Shape:")
print(Titanic.shape)

Dataset Shape:
(891, 12)

print("Dataset information:")
(Titanic.info())

Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

print("Descriptive Statistics:")
print (Titanic.describe())

Descriptive Statistics:
PassengerId  Survived  Pclass  Age  SibSp \
count  891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std    257.353842    0.486592    0.836071   14.526497    1.102743
min      1.000000    0.000000    1.000000    0.420000    0.000000
25%    223.500000    0.000000    2.000000   20.125000    0.000000
50%    446.000000    0.000000    3.000000   28.000000    0.000000
75%    668.500000    1.000000    3.000000   38.000000    1.000000
max    891.000000    1.000000    3.000000   80.000000    8.000000

Parch  Fare
count  891.000000  891.000000
```

mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

```
print("Checking for Null Values")
print(Titanic.isnull().sum())
```

```
Checking for Null Values
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64
```

✓ Data Cleaning

Missing values

Filling Missing age Values with Median

```
Titanic['Age'] = Titanic['Age'].fillna(Titanic['Age'].median())
```

```
Titanic['Embarked'] = Titanic['Embarked'].fillna('S')
```

```
Titanic['Cabin'] = Titanic['Cabin'].fillna('Unknown')
```

```
##Dropping unnecessary columns
```

```
Titanic = Titanic.drop(['PassengerId', 'Ticket'], axis=1)
```

```
# Creating a new feature: Family Size
```

```
Titanic['FamilySize'] = Titanic['SibSp'] + Titanic['Parch'] + 1 # +1 includes the passenger
```

```
# Creating a new feature: IsAlone
```

```
Titanic['IsAlone'] = 0
```

```
Titanic.loc[Titanic['FamilySize'] == 1, 'IsAlone'] = 1
```

```
## Title From Name
```

```
Titanic['Title'] = Titanic['Name'].str.extract('([A-Za-z]+)/.', expand = False)
```

```
## Group Rare Tiles
```

```
rare_titles = ['Lady', 'Countess', 'Capt', 'Col', 'Don', 'Dr', 'Major', 'Rev', 'Sir', 'Jonkheer', 'Dona']
```

```
Titanic.loc[Titanic['Title'].isin(rare_titles), 'Title'] = 'Rare'
```

```
Titanic.loc[Titanic['Title'] == 'Mlle', 'Title'] = 'Miss'
```

```
Titanic.loc[Titanic['Title'] == 'Ms', 'Title'] = 'Miss'
```

```
Titanic.loc[Titanic['Title'] == 'Mme', 'Title'] = 'Mrs'
```

```
## Categorical Features to Dummy Variables
```

```
Categorical_features = ['Pclass', 'Sex', 'Embarked', 'Title']
```

```
Titanic = pd.get_dummies(Titanic, columns=Categorical_features, drop_first=True)
```

```
print("After data cleaning and preparation:")
```

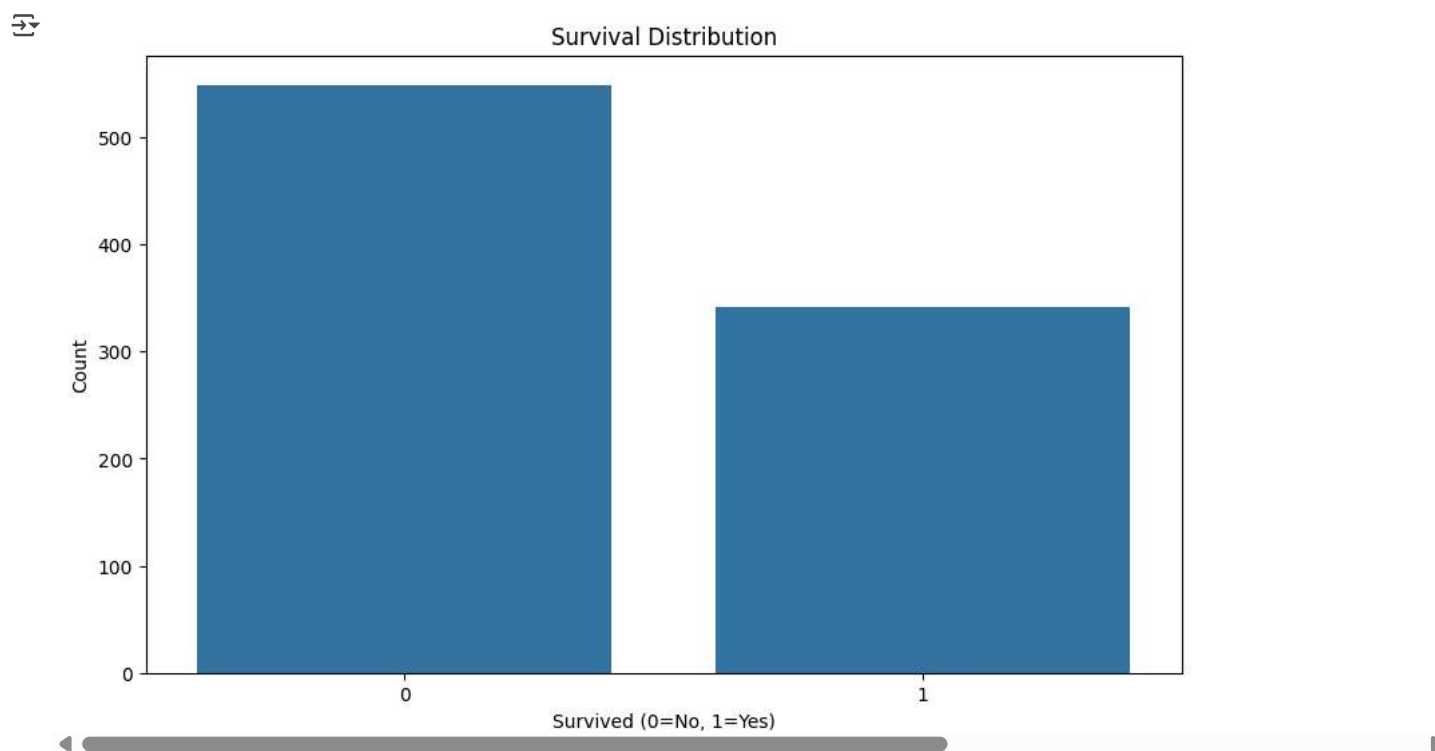
```
print(Titanic.isnull().sum())
```

↗ After data cleaning and preparation:

```
Survived    0
Name        0
Age         0
SibSp       0
Parch       0
Fare        0
Cabin       0
FamilySize  0
IsAlone     0
Pclass_2    0
Pclass_3    0
Sex_male    0
Embarked_Q  0
Embarked_S  0
dtype: int64
```

##Survival Distribution

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Survived', data=Titanic)
plt.title('Survival Distribution')
plt.xlabel('Survived (0=No, 1=Yes)')
plt.ylabel('Count')
plt.show()
```



Observation: Most passengers did not survive; the dataset is imbalanced with a higher number of non-survivors.

```
# Percentage of survivors
survival_rate = Titanic['Survived'].mean() * 100
print(f"Survival Rate: {survival_rate:.2f}%")
```

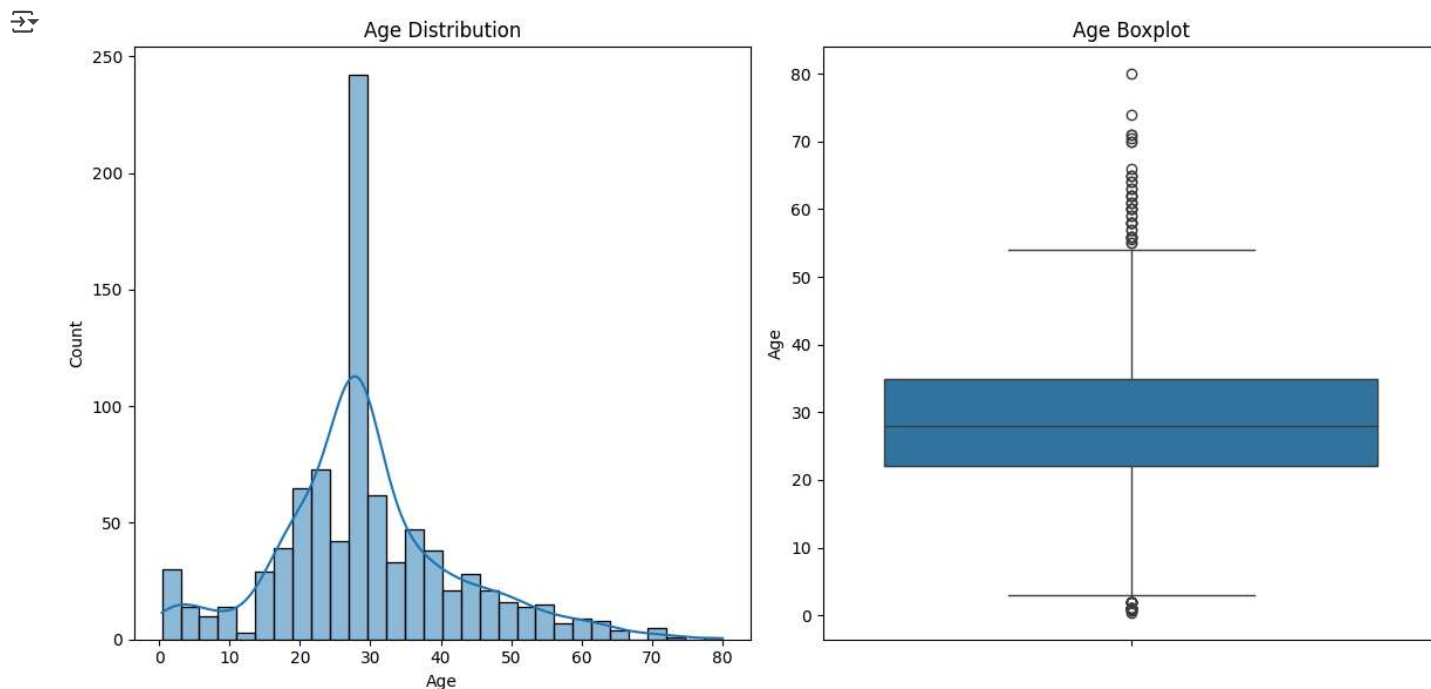
↗ Survival Rate: 38.38%

#Age Distribution

```
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1)
sns.histplot(Titanic['Age'], kde=True, bins=30)
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Count')
```

```
plt.subplot(1, 2, 2)
sns.boxplot(y=Titanic['Age'])
```

```
plt.title('Age Boxplot')
plt.ylabel('Age')
plt.tight_layout()
plt.show()
```



Observation: The majority of passengers were young adults, and there are few outliers as seen in the boxplot.

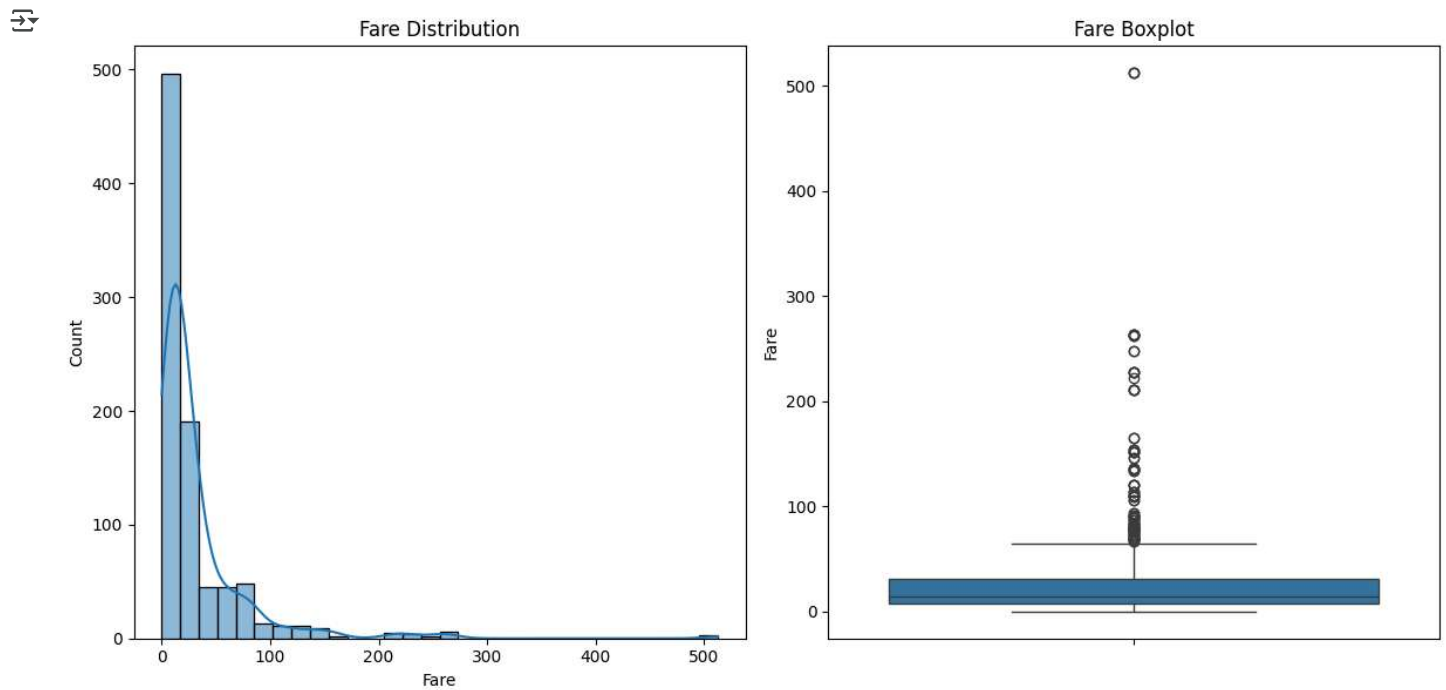
```
print(f"\nAge Statistics:")
print(Titanic['Age'].describe())
```

```
Age Statistics:
count      891.000000
mean       29.361582
std        13.019697
min         0.420000
25%        22.000000
50%        28.000000
75%        35.000000
max        80.000000
Name: Age, dtype: float64
```

Fare Distribution

```
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1)
sns.histplot(Titanic['Fare'], kde=True, bins=30)
plt.title('Fare Distribution')
plt.xlabel('Fare')
plt.ylabel('Count')

plt.subplot(1, 2, 2)
sns.boxplot(y=Titanic['Fare'])
plt.title('Fare Boxplot')
plt.ylabel('Fare')
plt.tight_layout()
plt.show()
```



Observation: Most passengers paid a lower fare, with a few high-paying outliers as shown in the right-skewed distribution.

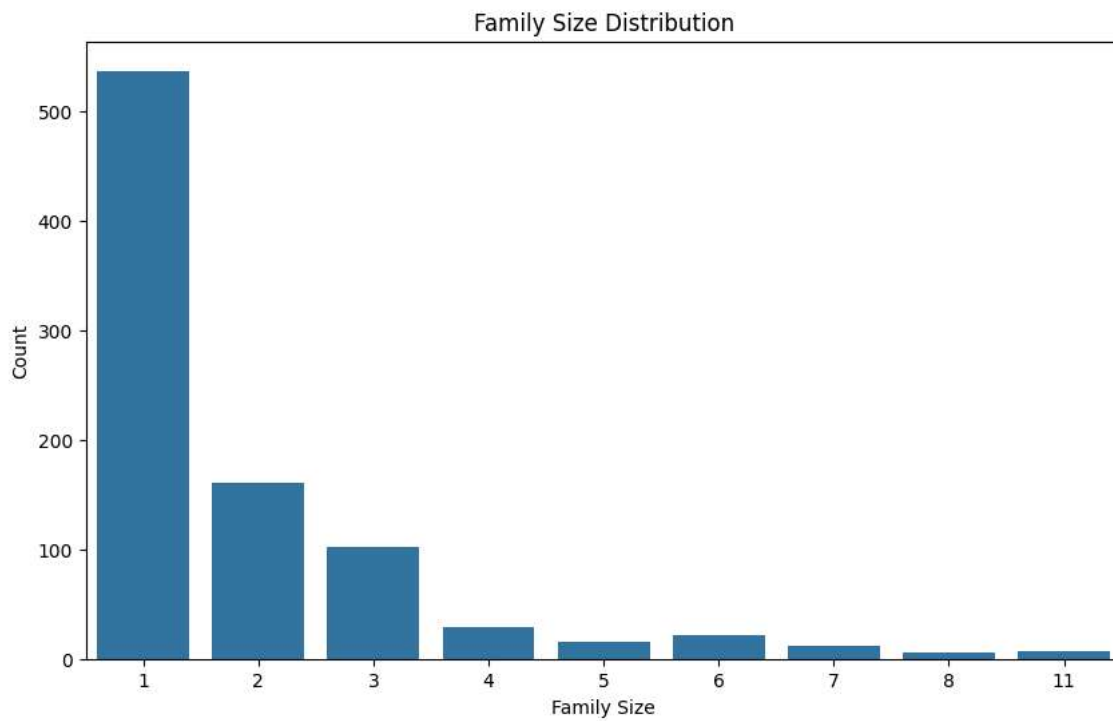
```
print(f"\nFare Statistics:")
print(Titanic['Fare'].describe())
```

```
Fare Statistics:
count      891.000000
mean       32.204208
std        49.693429
min         0.000000
25%         7.910400
50%        14.454200
75%        31.000000
max        512.329200
Name: Fare, dtype: float64
```

#Family Size Distribution

```
plt.figure(figsize=(10, 6))
sns.countplot(x='FamilySize', data=Titanic)
plt.title('Family Size Distribution')
plt.xlabel('Family Size')
plt.ylabel('Count')
plt.show()
```

```
print(f"\nFamily Size Statistics:")
print(Titanic['FamilySize'].value_counts())
```



Family Size Statistics:

FamilySize

1 537

2 161

3 102

4 29

6 22

5 15

7 12

11 7

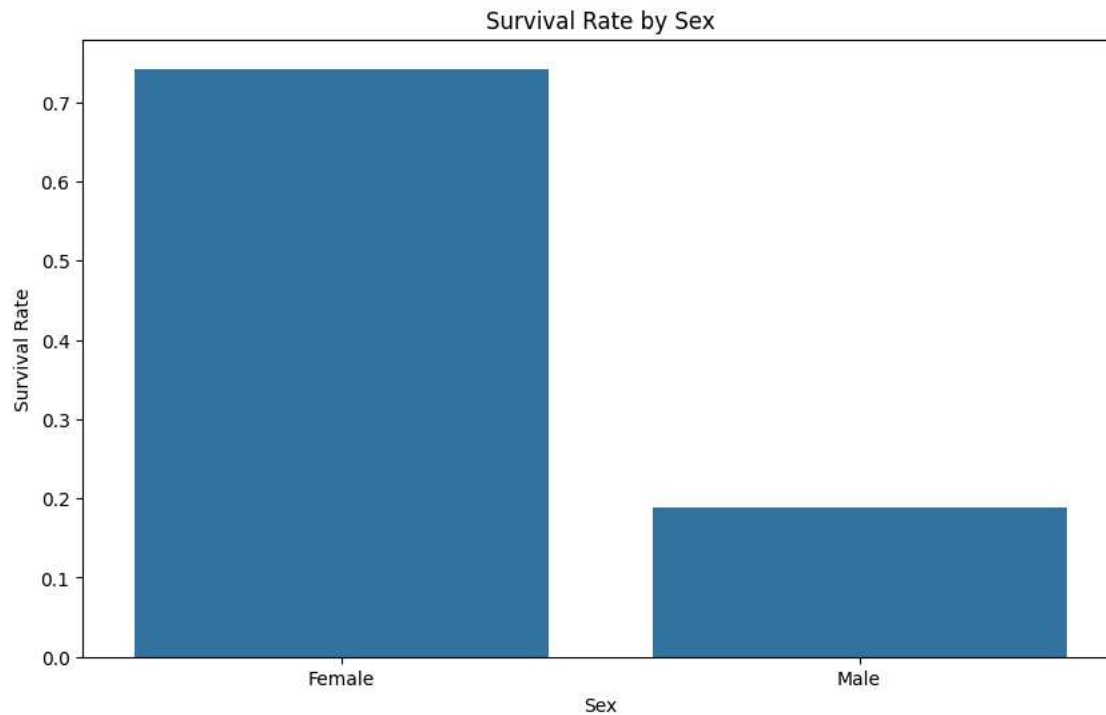
8 6

Name: count dtype: int64

Observation: The majority of passengers traveled alone or with small families; very large family groups were rare.

```
##Bivariate Analysis
# Survival by Sex
plt.figure(figsize=(10, 6))
sex_survival = Titanic.groupby(['Sex_male'])['Survived'].mean().reset_index()
sex_survival['Sex'] = ['Female', 'Male']
sns.barplot(x='Sex', y='Survived', data=sex_survival)
plt.title('Survival Rate by Sex')
plt.xlabel('Sex')
plt.ylabel('Survival Rate')
plt.show()

print("\nSurvival Rate by Sex:")
print(Titanic.groupby(['Sex_male'])['Survived'].mean())
```



Survival Rate by Sex:

Sex_male

False 0.742038

True 0.188908

Name: Survived, dtype: float64

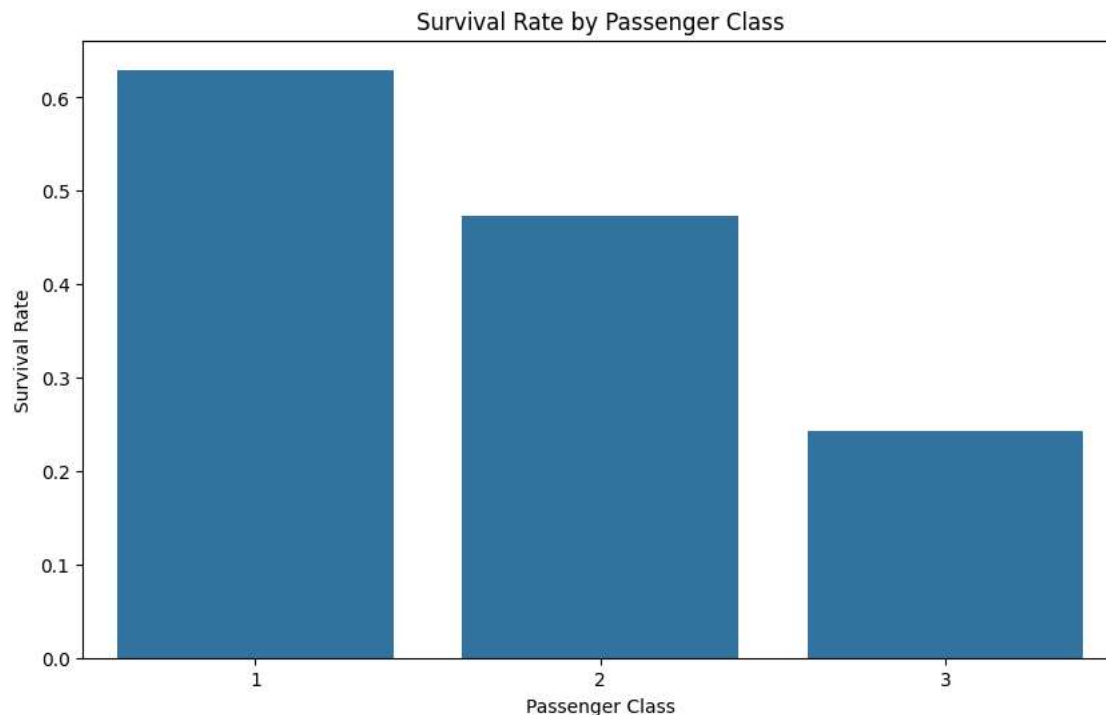
Observation: Females had a significantly higher survival rate compared to males.

Survival by Passenger Class

```
plt.figure(figsize=(10, 6))
# Create survival rates for each class
class1_survival = Titanic[Titanic['Pclass_2'] == 0][Titanic['Pclass_3'] == 0]['Survived'].mean()
class2_survival = Titanic[Titanic['Pclass_2'] == 1]['Survived'].mean()
class3_survival = Titanic[Titanic['Pclass_3'] == 1]['Survived'].mean()

pclass_survival = pd.DataFrame({
    'Pclass': [1, 2, 3],
    'Survival_Rate': [class1_survival, class2_survival, class3_survival]
})
sns.barplot(x='Pclass', y='Survival_Rate', data=pclass_survival)
plt.title('Survival Rate by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Survival Rate')
plt.show()
```

```
<ipython-input-184-4afebcdde66a>:5: UserWarning: Boolean Series key will be reindexed to match DataFrame index.
  class1_survival = Titanic[Titanic['Pclass_2'] == 0][Titanic['Pclass_3'] == 0]['Survived'].mean()
```



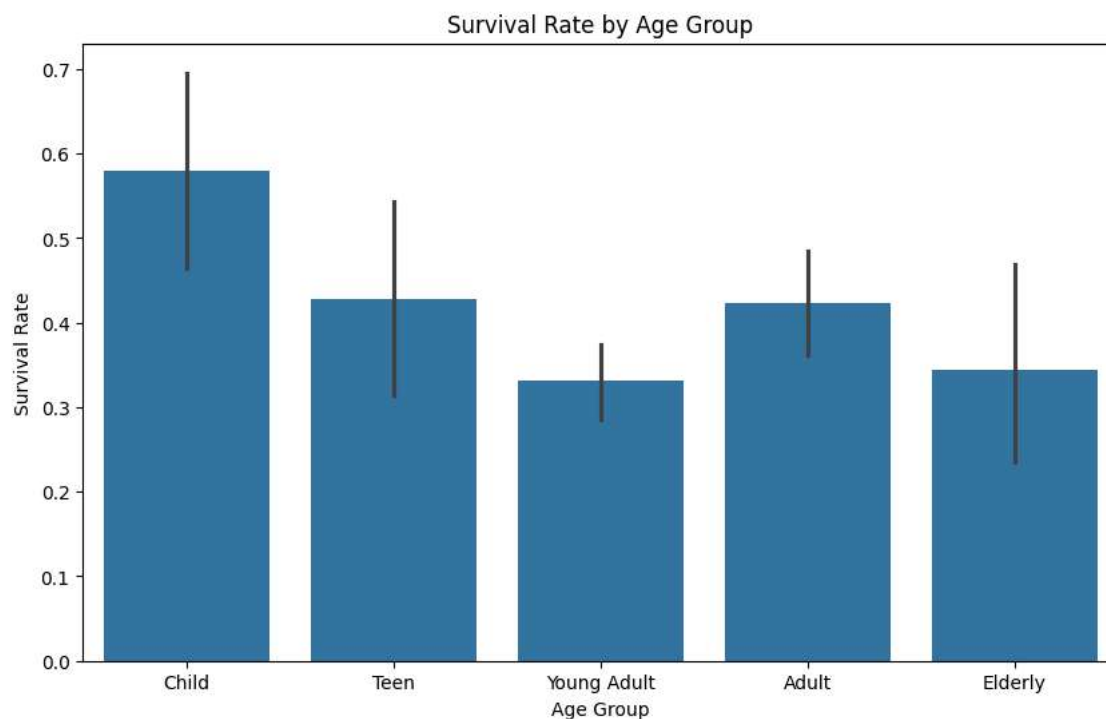
Observation: First-class passengers had the highest survival rate, while third-class had the lowest.

```
print("\nSurvival Rate by Passenger Class:")
print(pclass_survival)
```

```
Survival Rate by Passenger Class:
Pclass  Survival_Rate
0        1      0.629630
1        2      0.472826
2        3      0.242363
```

```
# Survival by Age
plt.figure(figsize=(10, 6))
age_bins = [0, 12, 18, 30, 50, 80]
age_labels = ['Child', 'Teen', 'Young Adult', 'Adult', 'Elderly']
Titanic['AgeGroup'] = pd.cut(Titanic['Age'], bins=age_bins, labels=age_labels)
sns.barplot(x='AgeGroup', y='Survived', data=Titanic)
plt.title('Survival Rate by Age Group')
plt.xlabel('Age Group')
plt.ylabel('Survival Rate')

plt.show()
```



Observation: Children had the highest survival rate, while survival among other age groups was more variable.

```
print("\nSurvival Rate by Age Group:")
print(Titanic.groupby('AgeGroup')['Survived'].mean())
```



Survival Rate by Age Group:

AgeGroup

Child 0.579710

Teen 0.428571

Young Adult 0.331096

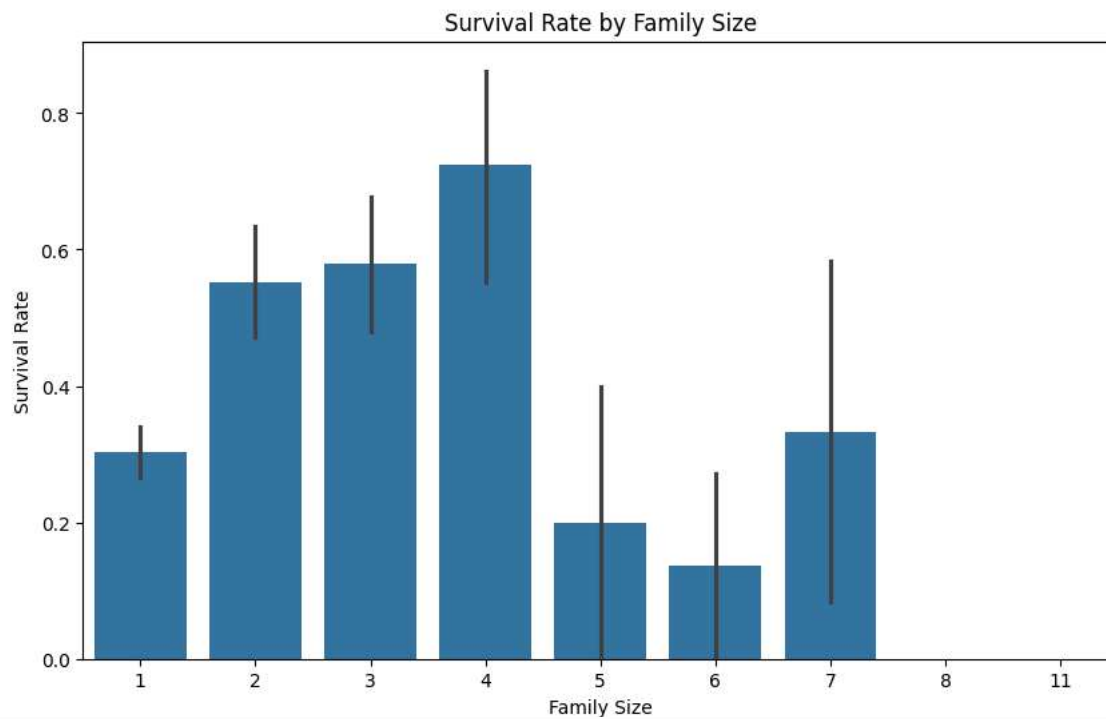
Adult 0.423237

Elderly 0.343750

Name: Survived, dtype: float64

<ipython-input-187-aaaa6ac7b092>:2: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future v
print(Titanic.groupby('AgeGroup')['Survived'].mean())

```
# Survival by Family Size
plt.figure(figsize=(10, 6))
sns.barplot(x='FamilySize', y='Survived', data=Titanic)
plt.title('Survival Rate by Family Size')
plt.xlabel('Family Size')
plt.ylabel('Survival Rate')
plt.show()
```



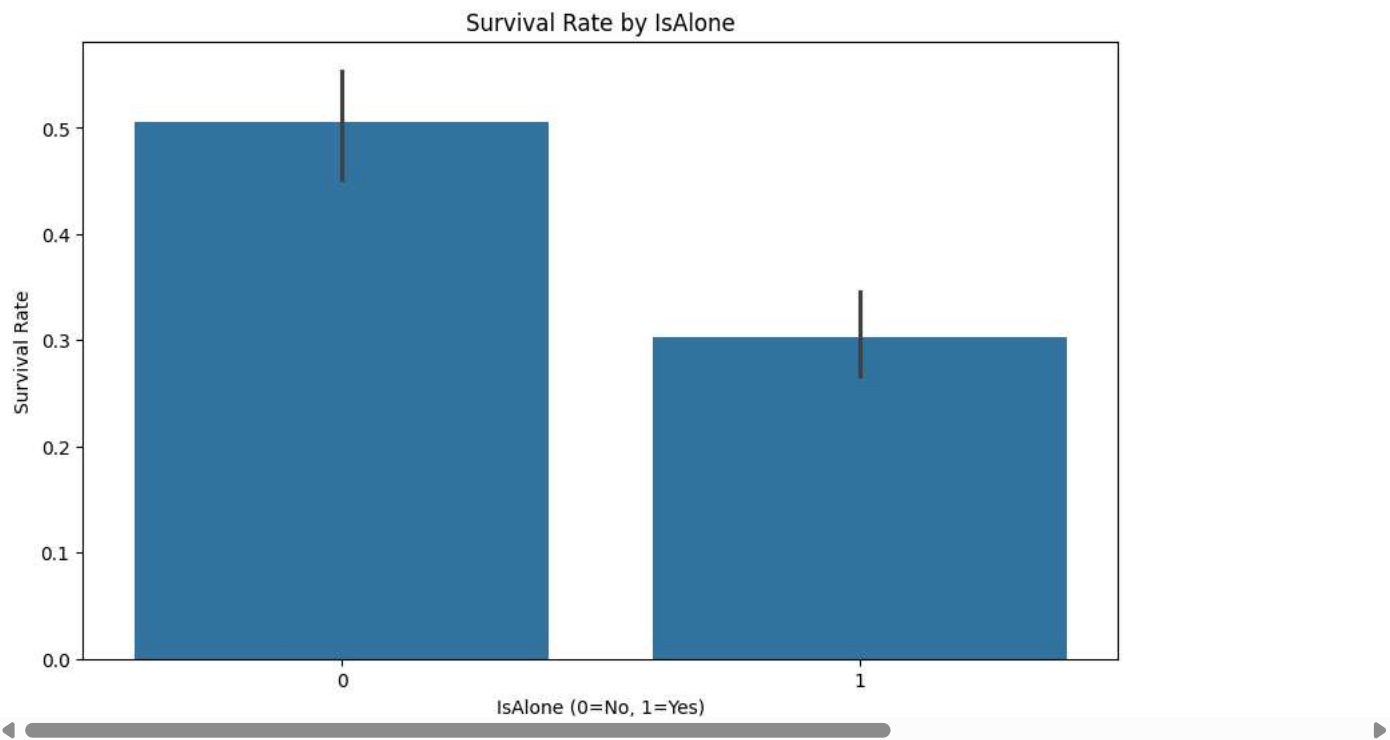
Observation: Passengers with small family sizes (2-4 members) had better survival chances than those alone or in large families.

```
print("\nSurvival Rate by Family Size:")
print(Titanic.groupby('FamilySize')['Survived'].mean())
```



```
Survival Rate by Family Size:
FamilySize
1    0.303538
2    0.552795
3    0.578431
4    0.724138
5    0.200000
6    0.136364
7    0.333333
8    0.000000
11   0.000000
Name: Survived, dtype: float64
```

```
# Survival by IsAlone
plt.figure(figsize=(10, 6))
sns.barplot(x='IsAlone', y='Survived', data=Titanic)
plt.title('Survival Rate by IsAlone')
plt.xlabel('IsAlone (0=No, 1=Yes)')
plt.ylabel('Survival Rate')
plt.show()
```



Observation: Passengers traveling alone had a lower survival rate than those accompanied by family.

```
print("\nSurvival Rate by IsAlone:")
print(Titanic.groupby('IsAlone')['Survived'].mean())
```



```
Survival Rate by IsAlone:
IsAlone
0    0.505650
1    0.303538
Name: Survived, dtype: float64
```

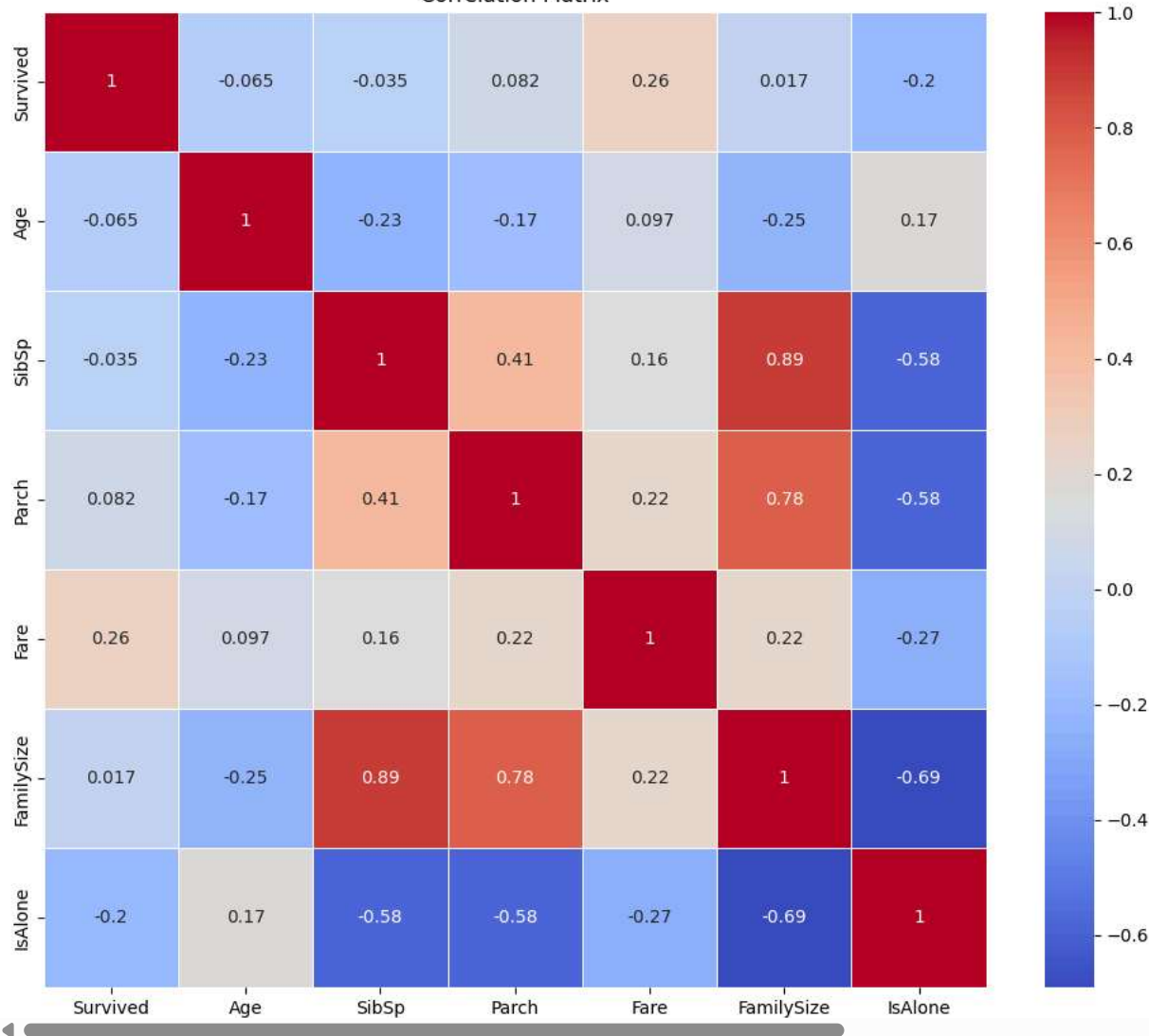
```
## Multivariate Analysis
```

```
# Correlation Matrix
```

```
plt.figure(figsize=(12, 10))
numerical_features = ['Survived', 'Age', 'SibSp', 'Parch', 'Fare', 'FamilySize', 'IsAlone']
correlation_matrix = Titanic[numerical_features].corr()
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', linewidths=0.5)
plt.title('Correlation Matrix')
plt.show()
```



Correlation Matrix



Observation: Fare, FamilySize, and IsAlone show moderate correlation with survival, while Age is weakly negatively correlated.

```
print("\nCorrelation Matrix:")
print(correlation_matrix)
```



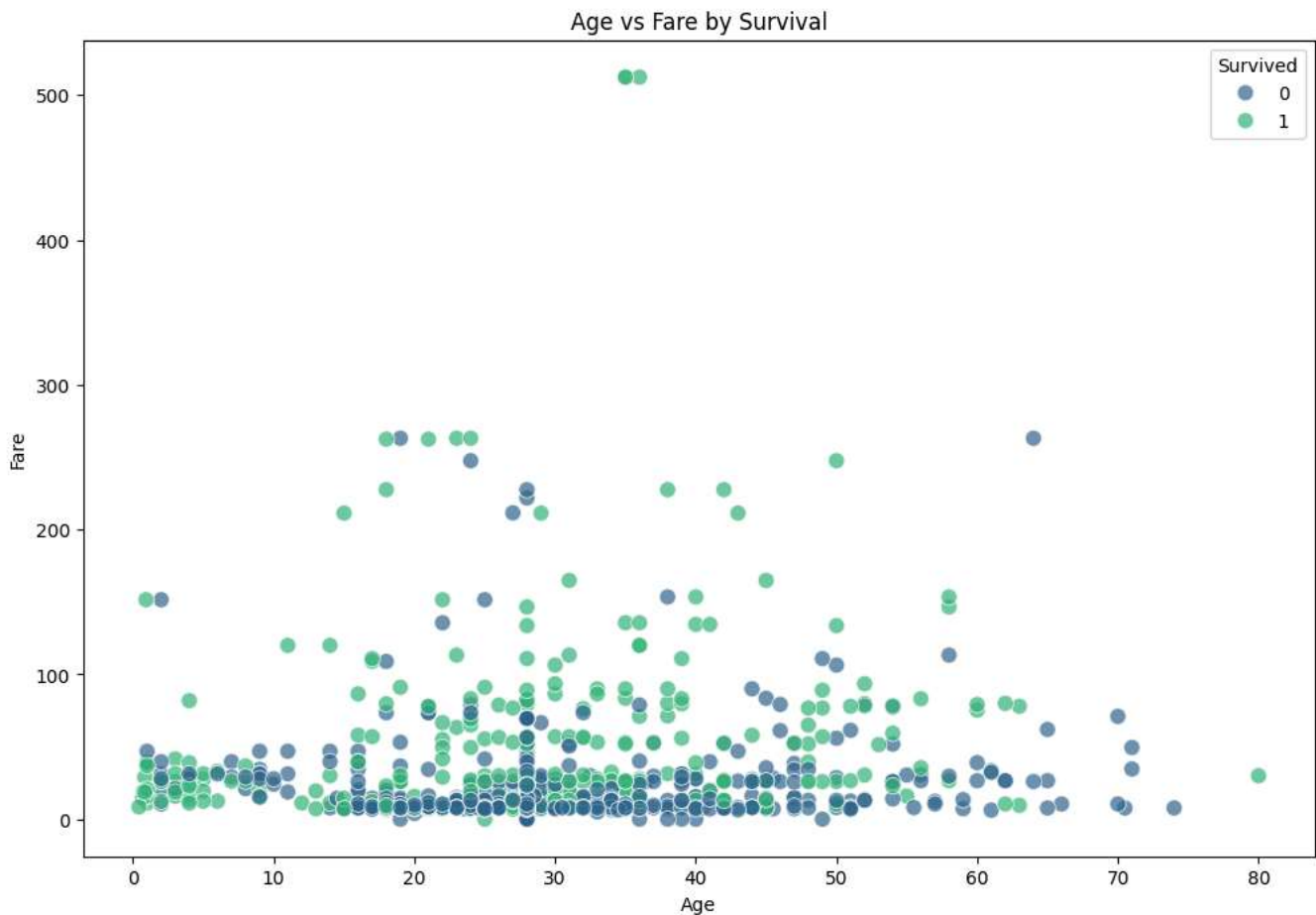
```
Correlation Matrix:
Survived    Survived    Age    SibSp    Parch    Fare    FamilySize  \
Survived    1.000000    -0.064910    -0.035322    0.081629    0.257307    0.016639
Age          -0.064910    1.000000    -0.233296    -0.172482    0.096688    -0.245619
SibSp        -0.035322    -0.233296    1.000000    0.414838    0.159651    0.890712
Parch         0.081629    -0.172482    0.414838    1.000000    0.216225    0.783111
Fare          0.257307    0.096688    0.159651    0.216225    1.000000    0.217138
FamilySize    0.016639    -0.245619    0.890712    0.783111    0.217138    1.000000
IsAlone       -0.203367    0.171647    -0.584471    -0.583398    -0.271832    -0.690922

IsAlone
Survived    -0.203367
Age           0.171647
SibSp        -0.584471
Parch        -0.583398
Fare         -0.271832
FamilySize   -0.690922
IsAlone       1.000000
```

```
# Age vs Fare by Survival
```

```
plt.figure(figsize=(12, 8))
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=Titanic, palette='viridis', s=80, alpha=0.7)
plt.title('Age vs Fare by Survival')
plt.xlabel('Age')
plt.ylabel('Fare')
```

```
plt.figure(figsize=(12, 8))
plt.legend(title='Survived', loc='upper right')
plt.show()
```



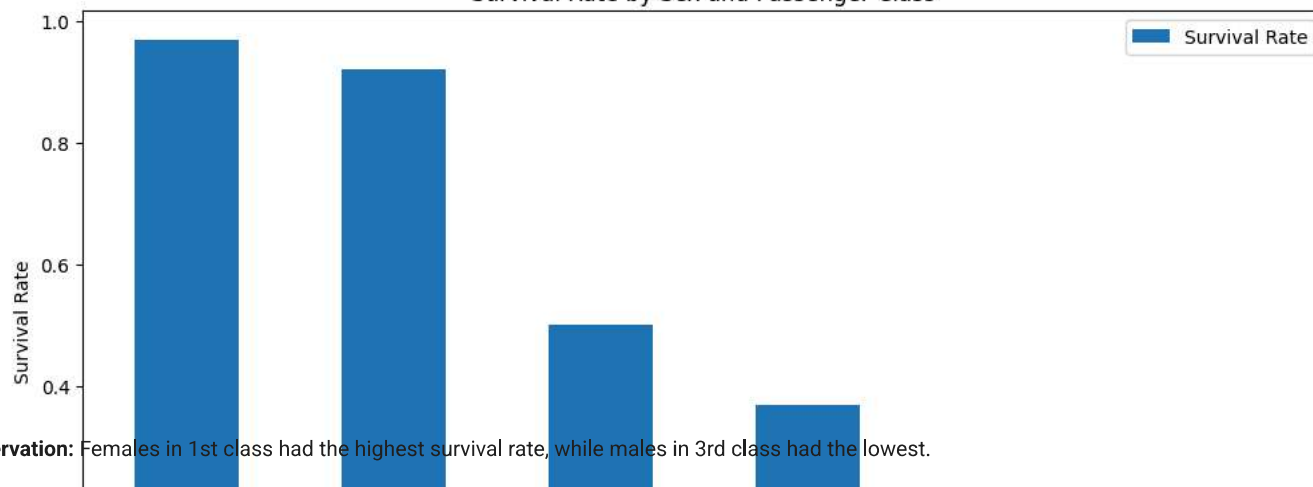
Observation: Survivors generally paid higher fares and were spread across various age groups.

Survival by Sex and Pclass

```
plt.figure(figsize=(12, 8))
sex_pclass_survival = pd.DataFrame({
    'Female_1st': Titanic[(Titanic['Sex_male'] == 0) & (Titanic['Pclass_2'] == 0) & (Titanic['Pclass_3'] == 0)]['Survived'].mean(),
    'Female_2nd': Titanic[(Titanic['Sex_male'] == 0) & (Titanic['Pclass_2'] == 1)]['Survived'].mean(),
    'Female_3rd': Titanic[(Titanic['Sex_male'] == 0) & (Titanic['Pclass_3'] == 1)]['Survived'].mean(),
    'Male_1st': Titanic[(Titanic['Sex_male'] == 1) & (Titanic['Pclass_2'] == 0) & (Titanic['Pclass_3'] == 0)]['Survived'].mean(),
    'Male_2nd': Titanic[(Titanic['Sex_male'] == 1) & (Titanic['Pclass_2'] == 1)]['Survived'].mean(),
    'Male_3rd': Titanic[(Titanic['Sex_male'] == 1) & (Titanic['Pclass_3'] == 1)]['Survived'].mean()
}, index=['Survival Rate']).T
sex_pclass_survival.plot(kind='bar', figsize=(12, 6))
plt.title('Survival Rate by Sex and Passenger Class')
plt.xlabel('Sex_Class')
plt.ylabel('Survival Rate')
plt.xticks(rotation=45)
plt.show()
```

<Figure size 1200x800 with 0 Axes>

Survival Rate by Sex and Passenger Class



Observation: Females in 1st class had the highest survival rate, while males in 3rd class had the lowest.

```
print("\nSurvival Rate by Sex and Passenger Class:")
print(sex_pclass_survival)
```



```
## Statistical Tests
# Survival vs Age
```

```
survived = Titanic[Titanic['Survived'] == 1]['Age']
not_survived = Titanic[Titanic['Survived'] == 0]['Age']
t_stat, p_value = stats.ttest_ind(survived, not_survived)
print("\nT-test for Age between Survivors and Non-survivors:")
print(f"t-statistic: {t_stat:.4f}")
print(f"p-value: {p_value:.4f}")
print(f"Conclusion: {'Statistically significant difference' if p_value < 0.05 else 'No statistically significant difference'}")
```

```
T-test for Age between Survivors and Non-survivors:
t-statistic: -1.9395
p-value: 0.0528
Conclusion: No statistically significant difference
```